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ASSIGNMENT – 03

Question :

Create a package named ticket_resolution on containing the following modules:

1. **Module 1 – ticket.py**
   - **Create a Ticket class to store ticket details.**

2. **Module 2 – ticketdao.py**
   - **Create a TicketDAO class to perform CRUD operations on tickets.**

3. **Module 3 – ticketui.py**
   - **Create a TicketUI class to provide a menu-based interface.**

4. **Main Script (main.py)**
   - **Invoke all operations using a menu-driven application.**

Answer:

Package Structure

ticket_resolution/

__init__.py

ticket.py

ticketdao.py

ticketui.py

main.py

Module 1: ticket.py

```
class Ticket:
def __init__(self, tid, issue, status):
self.tid = tid
self.issue = issue
```

```
self.status = status

def display(self):
print("ID:", self.tid)
print("Issue:", self.issue)
print("Status:", self.status)
print("------------------")
```

Module 2 – ticketdao.py

```
from ticket_resolution.ticket import Ticket

class TicketDAO:

  def __init__(self):

      self.tickets = []


  def add_ticket(self, ticket):

      self.tickets.append(ticket)

      print("Ticket added")


   def view_tickets(self):

      if not self.tickets:

         print("No tickets found")

      else:

         for t in self.tickets :

              t.display()


   def update_ticket(self, tid, new_status):

      for t in self.tickets:

         if t.tid == tid:

            t.status = new_status

            print("Ticket updated")
```

```
            return
  print("Ticket not found")


    def delete_ticket(self, tid):
        for t in self.tickets:
            if t.tid == tid:
                self.tickets.remove(t)
                print("Ticket deleted")
                return
        print("Ticket not found")
```

Module 3: ticketui.py

```
from ticket_resolution.ticket import Ticket
from ticket_resolution.ticketdao import TicketDAO
class TicketUI:
    def __init__(self):
        self.dao = TicketDAO()


  def menu(self):
        while True:
            print("\n1. Add Ticket")
            print("2. View Tickets")
            print("3. Update Ticket")
            print("4. Delete Ticket")
            print("5. Exit")
choice = int(input("Enter choice: "))
        if choice == 1:
```

```
            tid = int(input("Enter ID: "))
            issue = input("Enter issue: ")
            status = input("Enter status: ")
            t = Ticket(tid, issue, status)
            self.dao.add_ticket(t)
elif choice == 2:
            self.dao.view_tickets()
 elif choice == 3:
            tid = int(input("Enter ID: "))
            status = input("Enter new status: ")
            self.dao.update_ticket(tid, status)
elif choice == 4:
         tid = int(input("Enter ID: "))
         self.dao.delete_ticket(tid)
elif choice == 5:
       print("Exiting...")
       break
  else:
        print("Invalid choice")
```

Main Script (main.py)

```
from ticket_resolution.ticketui import TicketUI
def main():
      app = TicketUI()
      app.menu()
if __name__ == "__main__":
   main()
```

Output:

1. Add Ticket
2. View Tickets
3. Update Ticket
4. Delete Ticket
5. Exit